AKDENIZ UNIVERSITY · FACULTY OF ENGINEERING · DEPARTMENT OF COMPUTER ENGINEERING

CSE 491 — Project Proposal

Project title	NutriScan: Evidence-Based Food Ingredient and Health-Condition Matching
Group no.	9
Project supervisor	Dr. Taha Yiğit Alkan
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Group members

#	Name and surname	Student no.	Role / responsibility in the project
1	Levent Can Ceylan	20220808045	Knowledge-base schema and backend integration; LLM message layer; evaluation coordination; decision record editor
2	Şükran Hilal Hocaoğlu	20210808005	Health-condition list, guideline sources and rule table; evidence sources (CTD, JECFA); licence inventory; test sets
3	Ozan Karadaş	20220808007	Ingredient dictionary and matching pipeline (incl. vector search); Turkish terminology; mobile app and OCR

1. Summary

People with a health condition such as coeliac disease, diabetes, chronic kidney disease or asthma read food labels to decide whether a product suits them. The ingredient list is written in varying words ("E102", "tartrazin", "sarı renklendirici"), some terms are mixtures or classes rather than single substances, and the scientific sources that link substances to conditions are in English and rarely say how much matters. NutriScan is a mobile application that reads a packaged food's ingredient list and nutrition table with OCR, links each Turkish label phrase to a standard concept, and evaluates the product for the user's own conditions with a versioned, source-cited rule table. The result is one of four outcomes (not suitable, caution, no issue found, could not verify) shown with its rule, source and coverage; scientific associations (CTD, JECFA) are shown as research information and never turn into warnings by themselves. Matching runs in three steps: exact dictionary lookup, fuzzy matching, and vector search (Qdrant) as a fallback with an acceptance threshold. An LLM writes the final message from the structured result but cannot change it. By the end of the semester we deliver a working mobile prototype covering diet-relevant conditions with guideline-based rules, and a measured comparison of dictionary-only and hybrid matching on real Turkish labels.

2. Problem Definition and Motivation

Problem. Deciding whether a packaged food suits a person with a health condition requires three steps that are hard for a person and for software: recognising what each label phrase means, knowing which of those substances or nutrient amounts matter for the condition, and knowing how strong the evidence behind that link is. Label wording is inconsistent, Turkish terms rarely appear in international scientific databases, and associations in the literature do not state dose or causality.

Why it matters. Mistakes have a cost in both directions: missing a gluten source is a health issue for a person with coeliac disease, while warning about every ingredient that appears somewhere in the literature makes an application useless and scientifically wrong. Turkish labelling law makes the ingredient list and the main nutrition values mandatory [1], so the information needed for many decisions is on the package, but it is not machine-readable for the consumer.

Gap. Existing scanners score a product or apply diet filters, usually through barcodes and without showing why a warning was produced or how certain it is (Section 4). Large language models can describe a label photo, but they are not consistent across runs and do not separate a documented association from a guideline-based recommendation.

Target users. Adults in Türkiye managing a diet-relevant condition themselves (e.g., coeliac disease, diabetes, chronic kidney disease, hypertension, asthma with sulphite sensitivity) who read labels before buying.

3. Objectives, Scope and Success Criteria

Objectives

1. Read Turkish ingredient lists and nutrition tables from label photos (OCR), with user correction.

2. Split and type each label phrase (single chemical, chemical class, mixture, food raw material, nutrient value) and link it to a standard concept (E-number, CAS, MeSH) without inventing chemical identities.

3. Build a versioned rule table for diet-relevant conditions from clinical guidelines and food regulation, with three rule types: presence, amount, declaration.

4. Show each result with its rule, source, coverage and research evidence; say "could not verify" when data or rules are missing.

5. Measure the contribution of vector search over dictionary matching on real Turkish labels.

Out of scope

Barcode scanning and product databases; storing scanned products; product recommendations or alternatives (possible later feature); dose or exposure calculations; diagnosis, treatment advice or clinical validation; drug interactions and reconciliation of multiple conditions.

Success criteria

• Matching test (100 real label phrases with reference answers): dictionary-only and hybrid settings reported on accuracy, answer rate, precision of answered items and wrong-match count; hybrid kept only if it improves accuracy without raising wrong matches.

• Rule test (20 product × condition scenarios): zero missed "not suitable" outcomes; every rule row has a source, section, version and two independent reviewers.

• Same input and same rule version always give the same outcome.

• End-to-end demo on real Turkish labels on a phone, including the LLM message with template fallback.

4. Similar Systems and Background

System / work	What it does	Limitation or gap your project addresses
Yuka [2]	Barcode scan; 0–100 product score; premium diet filters	Single score, no health-condition rules; not available in Türkiye
Fig [3]	Barcode scan with thousands of diet and allergen filters	Filters without visible evidence or uncertainty; US market
Fooducate [4]	Product grades and tracking; premium modes for some conditions	US data; grades do not show rule source or coverage
Ürün Dedektörü [5]	Turkish barcode/label analysis with personal filters (allergy, pregnancy, coeliac)	Limited condition coverage (diabetes, kidney disease, asthma); no evidence level
CTD [6], WHO JECFA [7]	Curated chemical–disease associations; additive safety evaluations	Associations without dose or causality; English; not consumer rules
Biomedical entity linking (BioSyn [8], SapBERT [9])	Map free-text terms to standard concepts with lexical and dense retrieval	Evaluated on English biomedical text, not Turkish food labels

Our contribution is the combination: Turkish label-phrase linking with typed concepts, a rule table that separates guideline-based decisions from literature associations, and transparent results with coverage and uncertainty.

5. Preliminary Requirements

Functional requirements

ID	Requirement	Priority (must / should / could)
FR-1	The system shall let the user select one or more health conditions in a profile kept on the device.	must
FR-2	The system shall capture a label photo, extract the ingredient list and nutrition values with OCR, and let the user correct the text.	must
FR-3	The system shall split the ingredient list and assign each phrase a type (single chemical, chemical class, mixture, food raw material, nutrient, unknown).	must
FR-4	The system shall link phrases to canonical concepts by exact dictionary lookup, then fuzzy matching, then vector search with a threshold; unresolved phrases shall be marked "could not verify".	must
FR-5	The system shall evaluate the product for each selected condition with the versioned rule table and return one of four outcomes.	must
FR-6	The system shall show for each outcome the rule, its source and section, the data used and the coverage; CTD/JECFA evidence shall be shown as information only.	must
FR-7	The system shall generate the final user message with an LLM from the structured result, with a template fallback.	should
FR-8	The team shall be able to add aliases and rule rows through a reviewed data file, without code changes.	should
FR-9	The system could suggest products of the same category that better fit the profile.	could

Non-functional requirements

ID	Requirement	How it will be checked
NFR-1	Same input and same rule version produce the same outcome.	Repeated runs on the rule test set
NFR-2	The LLM message never changes the outcome or adds facts; no "safe/ harmful" wording.	Automated check of messages on the test set; banned-word list
NFR-3	No scanned product or health profile is stored on the server.	Schema review; request/log inspection
NFR-4	A result is shown within 3 s (p95) for a typical label after OCR.	Timing on the test set
NFR-5	Every rule row carries source, section, version and two reviewers.	Data validation script

6. Method and Technology Plan

Approach. Small, decision-first increments: one shared decision record defines the architecture; each component has one owner; results are measured on fixed test sets. Rules are data, so adding a condition adds rows, not code.

Architecture outline. Mobile app (Expo) captures the label and runs on-device OCR → Python service (FastAPI) splits and types phrases → matching pipeline: exact lookup and fuzzy matching on the alias table, then Qdrant nearest-neighbour search over alias embeddings with an acceptance threshold → rule engine evaluates presence, amount and declaration rules per condition (amounts come from the nutrition table, not from ingredient names) → evidence lookup adds CTD/JECFA information → an LLM turns the structured result into a short Turkish message; a checker rejects messages that change the outcome or use banned words, falling back to a template.

Data. A knowledge base only: concepts (E-number, CAS, MeSH, PubChem CID), aliases with relation types (same as, contextual, is-a, contains, may contain), conditions with coverage notes, rules (type, condition, outcome, missing-data outcome, source, section, version, two reviewers), evidence (CTD, JECFA, with study counts) and a source/licence inventory. Sources: Turkish Food Codex additive and labelling regulations [1][10], Open Food Facts ingredients taxonomy [11], CTD vocabulary and associations [6], PubChem [12], WHO JECFA [7], clinical guidelines such as ADA 2026 and KDIGO 2024 [13]

[14]. Only needed Turkish terms are added to the alias table, reviewed by two members; FooDB is excluded for licence reasons (CC BY-NC) [15].

Layer / component	Technology or tool	Reason for the choice
Mobile client and OCR	Expo (React Native), on-device text recognition (library chosen on 20 real labels)	One code base for iOS and Android; no photo upload
Matching and rule service	Python, FastAPI, rapidfuzz, multilingual embedding model	Text matching and embedding tools are Python-native
Knowledge base	PostgreSQL	Relational rules, sources and versions
Vector search	Qdrant (alias embeddings only)	Fallback matching step; contribution measured by switching it off
Final message	LLM with structured input and output checker	Readable explanations without letting the model decide

7. Work Plan

Week 1 = 14 September 2026; interim report around week 8, final report and demo around week 14.

WP	Work package and its output	Responsible member(s)	Weeks
WP1	Decision record, repository, knowledge-base schema, service skeleton	Levent	6–7
WP2	Ingredient dictionary v0: E-numbers, OFF taxonomy (Turkish), CTD vocabulary; reviewed Turkish aliases	Ozan	6–9
WP3	Condition list, guideline sources, rule table with two-person review; licence inventory	Hilal (review: Levent, Ozan)	6–12
WP4	Matching pipeline (split, type, exact, fuzzy, Qdrant) and the 100-phrase matching test	Ozan, Levent	7–13
WP5	Mobile app: OCR capture and correction, profile, result screen with sources and coverage; LLM message layer	Ozan (app), Levent (LLM, integration)	8–14
WP6	Rule test (20 scenarios), measurements, interim and final reports, TÜBİTAK 2209-A application	All (coordination: Levent)	8–14

Spring semester (CSE 492): wider condition coverage, curation tools, same-category alternatives, user study.

8. Expected Outputs and Risks

Expected outputs

A working mobile prototype that reads real Turkish labels and returns explained, source-cited results for diet-relevant conditions; the knowledge base (dictionary, rule table, evidence links, source inventory); the matching and rule test sets with measured results; the repository with documentation; interim and final reports; a TÜBİTAK 2209-A application.

Risks

Risk	Likelihood	Impact	Mitigation plan
OCR errors on curved or glossy packages	High	Medium	User correction step; fuzzy matching; library chosen on real labels
Turkish terms missing from dictionaries	High	Medium	E-number bridge; reviewed team aliases; vector fallback with threshold; "could not verify"
Guideline misread without a clinician	Medium	High	Two independent reviewers per rule; source section stored; claim limited to guideline conformity
LLM message adds or changes facts	Medium	High	Structured input only; output checker; template fallback

Vector search adds little over dictionary	Medium	Low	Measured result is reported either way; component can be removed
Data licences (FooDB, OFF)	Medium	Medium	FooDB excluded; OFF used with attribution; source inventory
Medical-device boundary [16]	Low	High	Informational wording, no diagnosis or treatment claims, in-app disclaimer

9. References

[1] Türk Gıda Kodeksi Gıda Etiketleme ve Tüketicileri Bilgilendirme Yönetmeliği, Resmî Gazete 29960 (mükerrer), 26.01.2017.

[2] Yuka, help centre, https://help.yuka.io

[3] Fig, https://foodisgood.com/

[4] Fooducate, https://www.fooducate.com/

[5] Ürün Dedektörü, https://urundedektoru.com/

[6] A. P. Davis et al., "Comparative Toxicogenomics Database's 20th anniversary: update 2025," Nucleic Acids Research, vol. 53, no. D1, pp. D1328–D1334, 2025.

[7] WHO, "Evaluations of the Joint FAO/WHO Expert Committee on Food Additives (JECFA)," https://apps.who.int/food-additives-contaminants-jecfa-database/

[8] M. Sung, H. Jeon, J. Lee and J. Kang, "Biomedical Entity Representations with Synonym Marginalization," in Proc. 58th Annual Meeting of the ACL, 2020, pp. 3641–3650.

[9] F. Liu, E. Shareghi, Z. Meng, M. Basaldella and N. Collier, "Self-Alignment Pretraining for Biomedical Entity Representations," in Proc. NAACL-HLT, 2021, pp. 4228–4238.

[10] Türk Gıda Kodeksi Gıda Katkı Maddeleri Yönetmeliği, Resmî Gazete 28693, 30.06.2013 (with later amendments).

[11] Open Food Facts, ingredients taxonomy, https://github.com/openfoodfacts/openfoodfacts-server/blob/main/taxonomies/food/ingredients.txt

[12] NCBI PubChem, https://pubchem.ncbi.nlm.nih.gov/

[13] American Diabetes Association Professional Practice Committee, "5. Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes: Standards of Care in Diabetes—2026," Diabetes Care, vol. 49, suppl. 1, pp. S89–S131, 2026, doi:10.2337/dc26-S005.

[14] KDIGO CKD Work Group, "KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease," Kidney International, vol. 105, no. 4S, pp. S117–S314, 2024, doi:10.1016/j.kint.2023.10.018.

[15] FooDB Version 1.0, licence statement (CC BY-NC 4.0), https://foodb.ca/

[16] TİTCK, Tıbbi Cihaz Yönetmeliği, Resmî Gazete 31499 (mükerrer), 02.06.2021; Regulation (EU) 2017/745 on medical devices.

Supervisor Approval

To be completed by the project supervisor. The proposal is submitted to MS Teams after the supervisor has reviewed it.

Supervisor name	
Assessment	Approved ☐ Approved with revisions ☐ Revise and resubmit ☐
Comments	
Date / signature	